

Greeks

Risk Alpha

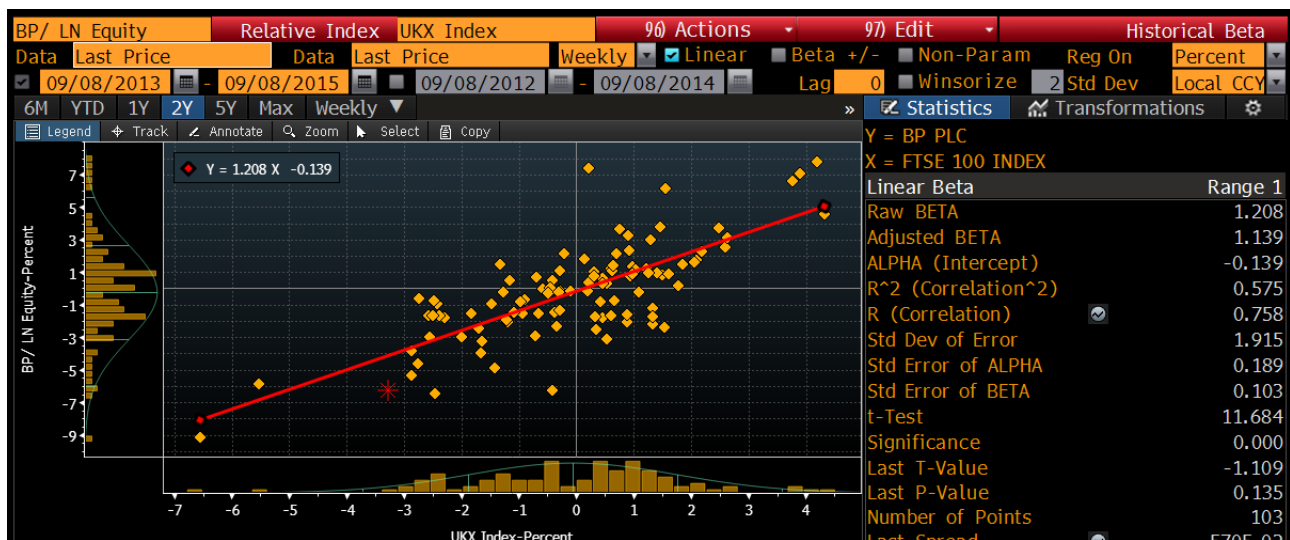
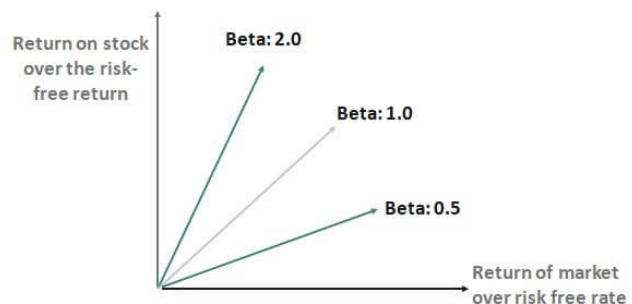
Alpha represents the excess return of an investment or portfolio compared to a benchmark. It measures the performance of an investment relative to the market or a chosen benchmark. Positive alpha indicates outperformance, while negative alpha suggests underperformance. Do not consider the alpha as an error term.

$$\mathbb{E}(asset) = r_f + \beta_{asset}(r_{market} - r_f) + \alpha + \epsilon_{asset}$$

Risk Beta

Beta is a measure of systematic risk of a particular asset or investment in relation to the overall market. It indicates how sensitive the asset's returns are to fluctuations in the market. A beta of 1 means the asset tends to move in line with the market, while a beta greater than 1 implies the asset is more volatile than the market. Conversely, a beta less than 1 suggests the asset is less volatile than the market.

$$\beta_{asset} = \frac{cov(R_{asset}, R_{market})}{\sigma_{market}^2}$$



In general, to change the beta of the portfolio from b to b^* , where $b \neq b^*$, a short position in

$$(\beta_0 - \beta_1) \frac{V_A}{V_F}$$

contracts is required. When $b \leq b^*$, a long position in

$$(\beta_1 - \beta_0) \frac{V_A}{V_F}$$

contracts is required.

Risk Delta

In the context of finance, delta (Δ) refers to the sensitivity of an option's price to changes in the price of the underlying asset. It measures the degree to which the option price moves in relation to the underlying asset's price movement. Delta can range from 0 to 1 for call options (0 to -1 for put options), where a higher delta indicates a stronger correlation between the option price and the underlying asset's price. A rise in the price of the underlying stock is positive for call options but not for put options. This explains that the Delta value is positive for call options and negative for put options.

$$\Delta = \frac{\partial V}{\partial S} \quad (= V_S)$$

For options traders, delta also represents the hedge ration for creating a delta-neutral position.

For example, if you purchase a standard American call option with a 0.40 delta, you will need to sell 40 shares of stock to be fully hedged. Net delta for a portfolio of options can also be used to obtain the portfolio's hedge ratio.

It can be shown that, under the Black-Scholes framework, the delta for a call and a put option are given by

$$\begin{aligned}\Delta_c &= N(d_1) \cdot e^{-\delta T} \\ \Delta_p &= -N(-d_1) \cdot e^{-\delta T}\end{aligned}$$

Risk Gamma

Gamma (Γ) represents the rate of change in an option's delta with respect to changes in the price of the underlying asset, or the second partial derivative of the value of the asset with respect to the stock price. It measures how quickly the sensitivity of the option's price changes as the underlying asset's price moves.

$$\Gamma = \frac{\partial \Delta}{\partial S} = \frac{\partial^2 V}{\partial S^2} \quad (= V_{SS})$$

Gamma indicates the amount the delta would change given a \$1 move in the underlying security. It is used to determine how stable an option's delta is: Higher gamma values indicate that delta could change dramatically in response to even small movements in the underlying's price. Gamma values are generally smaller the further away from the date of expiration; options with longer expirations are less sensitive to delta changes. As expiration approaches, gamma values are typically larger, as price changes have more impact on gamma.

Options traders may opt to not only hedge delta but also gamma in order to be delta-gamma-neutral, meaning that as the underlying price moves, the delta will remain close to zero.

For example, assume an investor is long on a call option. The call option has a delta of 0.50 and a gamma of 0.10. Therefore, if stock increases or decreases by \$1, the call option's delta would increase or decrease by 0.10.

It can be shown that, under the Black-Scholes framework, the delta for a call and a put option are given by

$$\Gamma = \frac{1}{S \cdot \sigma \cdot \sqrt{2\pi T}} e^{-0.5 \delta T d_1^2}$$

Risk Theta

Theta (Θ) measures the rate of decline in the value of an option as time passes, or time sensitivity, assuming other factors remain constant. It quantifies the time decay of an option and reflects the erosion of its value over time.

$$\Theta = \frac{\partial V}{\partial t} \quad (= V_t)$$

Theta indicates the amount an option's price would decrease as the time to expiration decreases, all else equal. Options closer to expiration also have accelerating time decay. Long calls and long puts will usually have negative theta; short calls and short puts will have positive theta. By comparison, an instrument whose value is not eroded by time, such as a stock, would have zero theta.

For example, assume an investor is long an option with a theta of -0.50. The option's price would decrease by 50 cents every day that passes, all else being equal.

Risk Vega

Vega (ν) represents the sensitivity of an option's price to changes in the implied volatility of the underlying asset. It measures how the option price changes with fluctuations in market volatility. High vega indicates greater sensitivity to volatility changes.

$$\nu = \frac{\partial V}{\partial \sigma} \quad (= V_\sigma)$$

It represents the rate of change between an option's value and the underlying asset's implied volatility. This is the option's sensitivity to volatility. Vega indicates the amount an option's price changes given a 1% change in implied volatility.

Because increased volatility implies that the underlying instrument is more likely to experience extreme values, a rise in volatility will correspondingly increase the value of an option. Conversely, a decrease in volatility will negatively affect the value of the option. Vega is at its maximum for at-the-money options that have longer times until expiration.

For example, an option with a vega of 0.10 indicates the option's value is expected to change by 10 cents if the implied volatility changes by 1%.

Risk Rho & Psi

Rho (ρ) represents the sensitivity of an option's price to changes in the risk free rate. It measures how the option price changes with fluctuations in risk free rates.

$$\rho = \frac{\partial V}{\partial r_f} \quad (= V_r)$$

Psi (Ψ) represents the sensitivity of an option's price to changes in the dividend rate. It measures how the option price changes with fluctuations in dividends.

$$\Psi = \frac{\partial V}{\partial \delta} \quad (= V_\delta)$$